

Wenjie Luo

Data Analyst | Decision Optimization | Graduating Jan 2027

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PROFILE

Turn business questions into traceable metrics and analytical models. Experience in market research, SQL, mathematical programming and transport data, using controlled comparisons to support decisions.

EDUCATION

- National University of Singapore | MSc Civil Engineering** Aug 2025 - Jan 2027
Transport focus; research in transportation information and control.
- Chang'an University | BEng Transportation Engineering** Sep 2021 - Jun 2025
GPA 3.16/5.0; approximately top 15% of the major.

EXPERIENCE

- Chery International | Product Intern** May 2026 - Jun 2026
 - Combined competitor data, registrations and customer feedback into 15 personas and 5 VOC reports; used length, price and registration bubble charts for product positioning.
 - Produced sales-trend forecasts and mapped KD production-to-logistics activities into a Gantt plan to support market analysis and cross-functional scheduling.
- Expeditors | Supply Chain Assistant** Dec 2024 - Apr 2025
 - Improved document-processing spreadsheets with VLOOKUP and pivot tables, increasing efficiency by approximately 30%; refined existing standardized checks to reduce manual reconciliation errors.

PROJECTS

- Malaysia Automotive Metrics and Data Assistant** Aug 2026
 - Separated wholesale, registrations and share through metric definitions and SQL templates, with 14 nodes and 20 edges linking outputs to upstream tables in a demonstration warehouse.
 - On 30 authored questions, 22 returned correct results and 8 requested clarification. Tested guard behavior through ablation and documented the channel-turnover assumptions used to derive registrations.
- LP/MIP Modeling and Solver Experiments** Sep 2026
 - Built modeling, two-phase simplex and branch-and-bound; checked 11 instances against HiGHS. The same vehicle schedule needed 1 node with assignment modeling versus 21–31 with big-M.
 - Validated pure-integer Gomory cuts by enumerating 5,376 binary points. Three knapsacks fell to 1 search node; retained a 29-to-31-node counterexample to quantify cut overhead.
- NAVSIM | Factorial Experiments and Evaluation** Aug - Sep 2026
 - Evaluated 12,146 scenes with a common cache, a map-by-agent 2x2 input design and three seeds per cell; estimated paired 95% bootstrap intervals per comparison.
 - Map information added 0.296 PDMS versus 0.040 for ground-truth boxes. Adding boxes to map-aware models significantly reduced scores across all three seeds while open-loop L1 barely changed.

SKILLS

Analytics: Python, SQL, Excel, metric dictionaries, table lineage, ArcGIS, DBSCAN
Methods: LP/MIP, two-phase simplex, branch-and-bound, paired bootstrap, factorial experiments